



Santa In The Summer

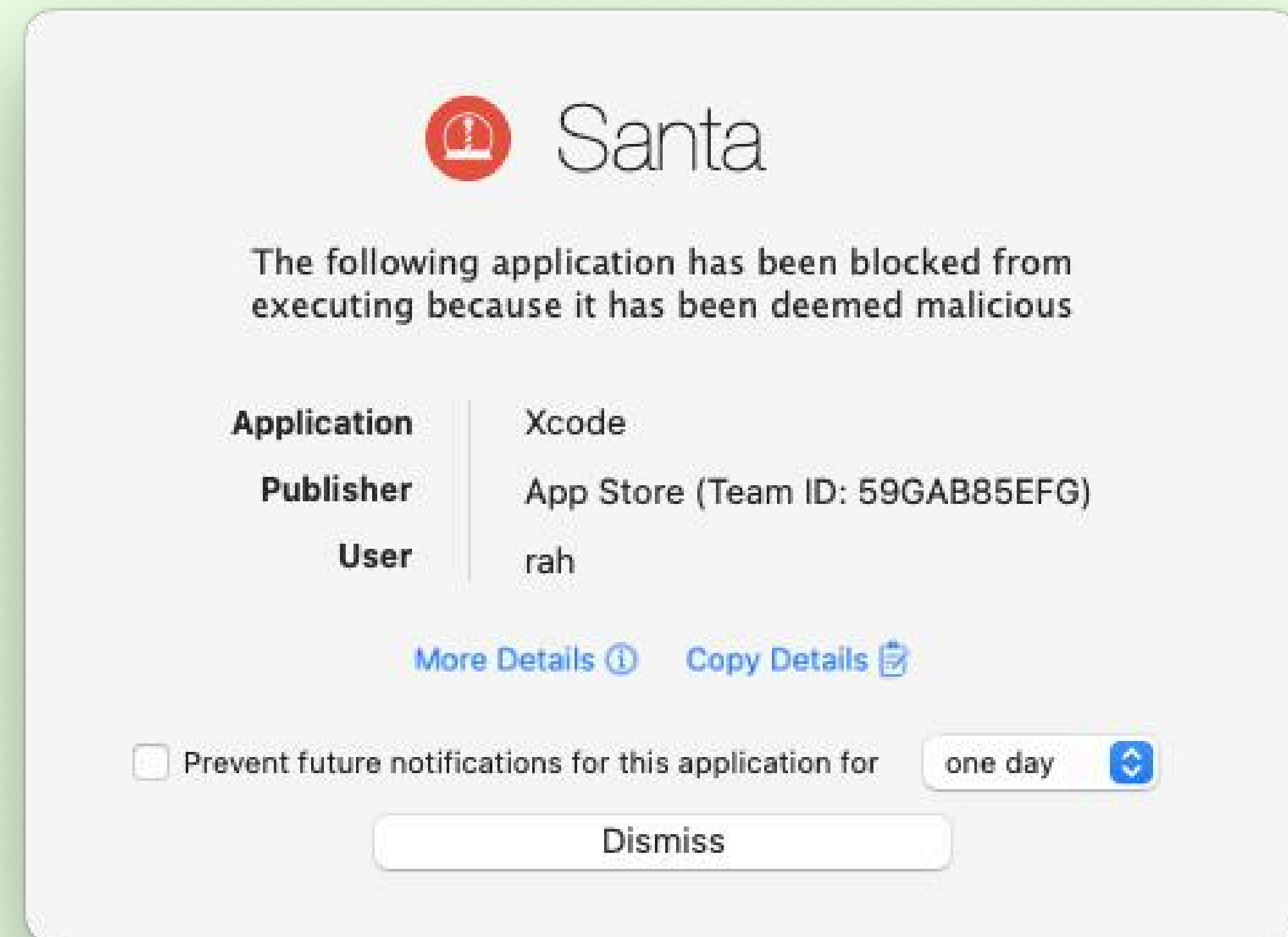
Russell Hancox

northpole.security

WHAT IS SANTA?

Santa is an open-source security agent for macOS.

It handles binary authorization, file-access authorization, USB mass storage authorization, and voluminous security event logging.



INTRO

WE ARE NORTH POLE SECURITY

Founded in 2024 by the four
primary contributors to Santa, our
company goal is to bring usable
guard-rail security to everyone



Sleighing Malware

Since 2024

SANTA IN THE SUMMER

SANTA STAYS OPEN- SOURCE

North Pole Security continues to develop Santa in the open, on GitHub.

WE WELCOME ISSUES AND PRS

We provide support in the #santa channel on MacAdmins Slack.

NEW FEATURES

**SINCE WE
OPENED IN
OCTOBER**

8

Releases

NEW FEATURES

**SINCE WE
OPENED IN
OCTOBER**

347

Merged PRs

SO FAR, WE'VE ADDED

FAA

Process-centric Rules
Signing ID Wildcards

Telemetry

TCC
XProtect
Clone/Copyfile
Gatekeeper Override
Authentication

Sync

Signing Status
Entitlements
SIP Status

santactl

fileinfo lists entitlements
doctor
eventupload

GUI

New interface
Translations
Sync Button
Customizable Silence Periods
Drag & Drop File Info

Standalone Mode

Tamper Protection

New Docs

Docs Search

Config Generator

Config

Telemetry type control
StaticRules validation

LET'S DIG IN

 Santa

Access to a file has been denied

Path Accessed	/Users/rah/Library/Applic...le/Chrome/Default/Cookies
Application	sqlite3
User	rah
Rule Name	ChromeCookies
Rule Version	v1.1

[More Details ⓘ](#)

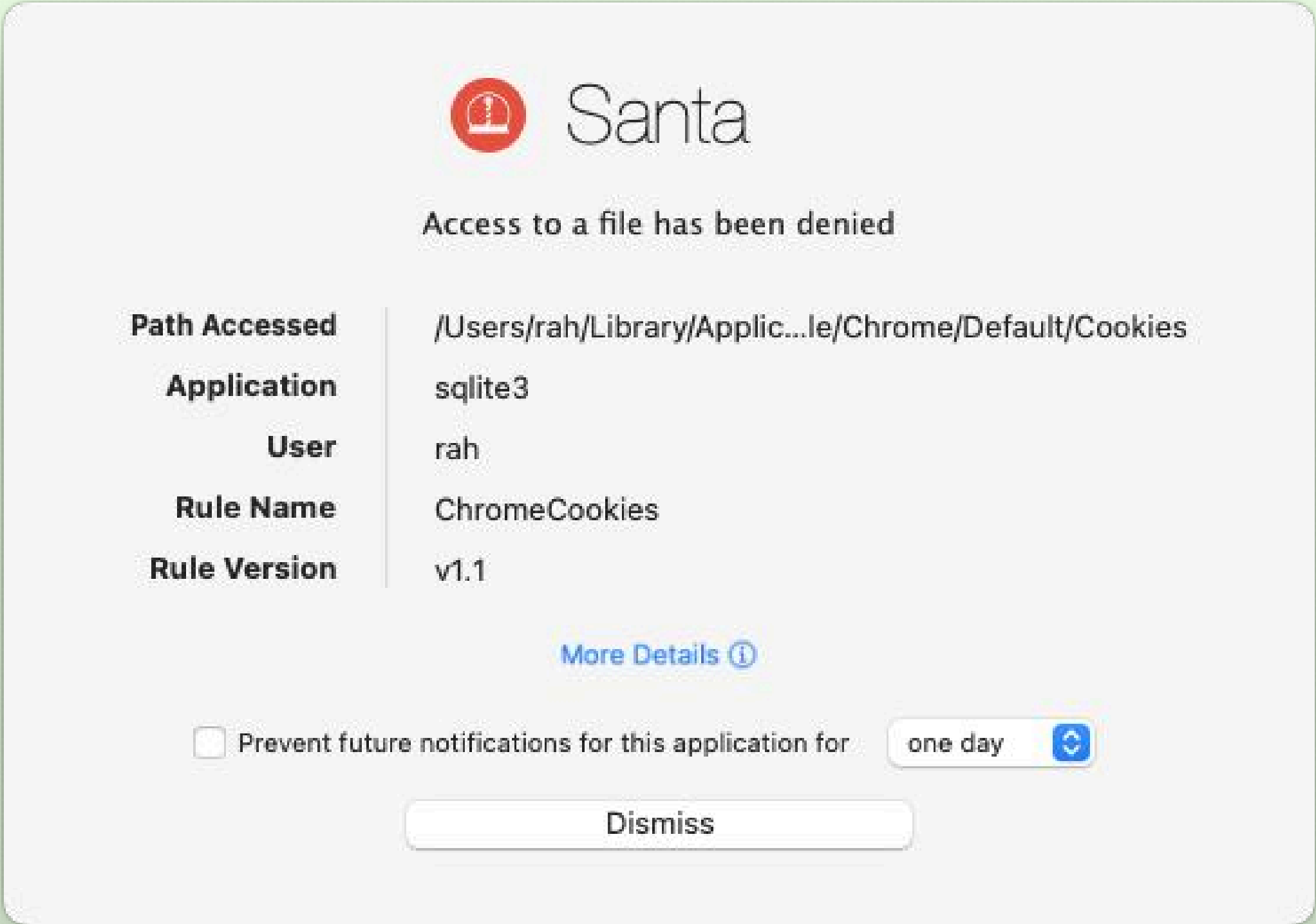
☐ Prevent future notifications for this application for

one day 

Dismiss

FAA

File-access Authorization is not a new feature in Santa, but we have been adding significant features and there's more to come





Protecting browser cookies

Keep those SaaS credentials
secure!

```
<key>ChromeCookies</key>
<dict>
  <key>Paths</key>
  <array>
    <dict>
      <key>Path</key>
      <string>/Users/*/Library/Application Support/Google/Chrome/*/Cookies</string>
      <key>IsPrefix</key>
      <true/>
    </dict>
  </array>
  <key>Options</key>
  <dict>
    <key>AllowReadAccess</key>
    <false/>
    <key>AuditOnly</key>
    <false/>
    <key>RuleType</key>
    <string>PathsWithAllowedProcesses</string>
  </dict>
  <key>Processes</key>
  <array>
    <dict>
      <key>SigningID</key>
      <string>com.google.Chrome*</string>
      <key>TeamID</key>
      <string>EQHXZ8M8AV</string>
    </dict>
    <dict>
      <key>SigningID</key>
      <string>com.apple.mdworker_shared</string>
      <key>PlatformBinary</key>
      <true/>
    </dict>
  </array>
</dict>
```



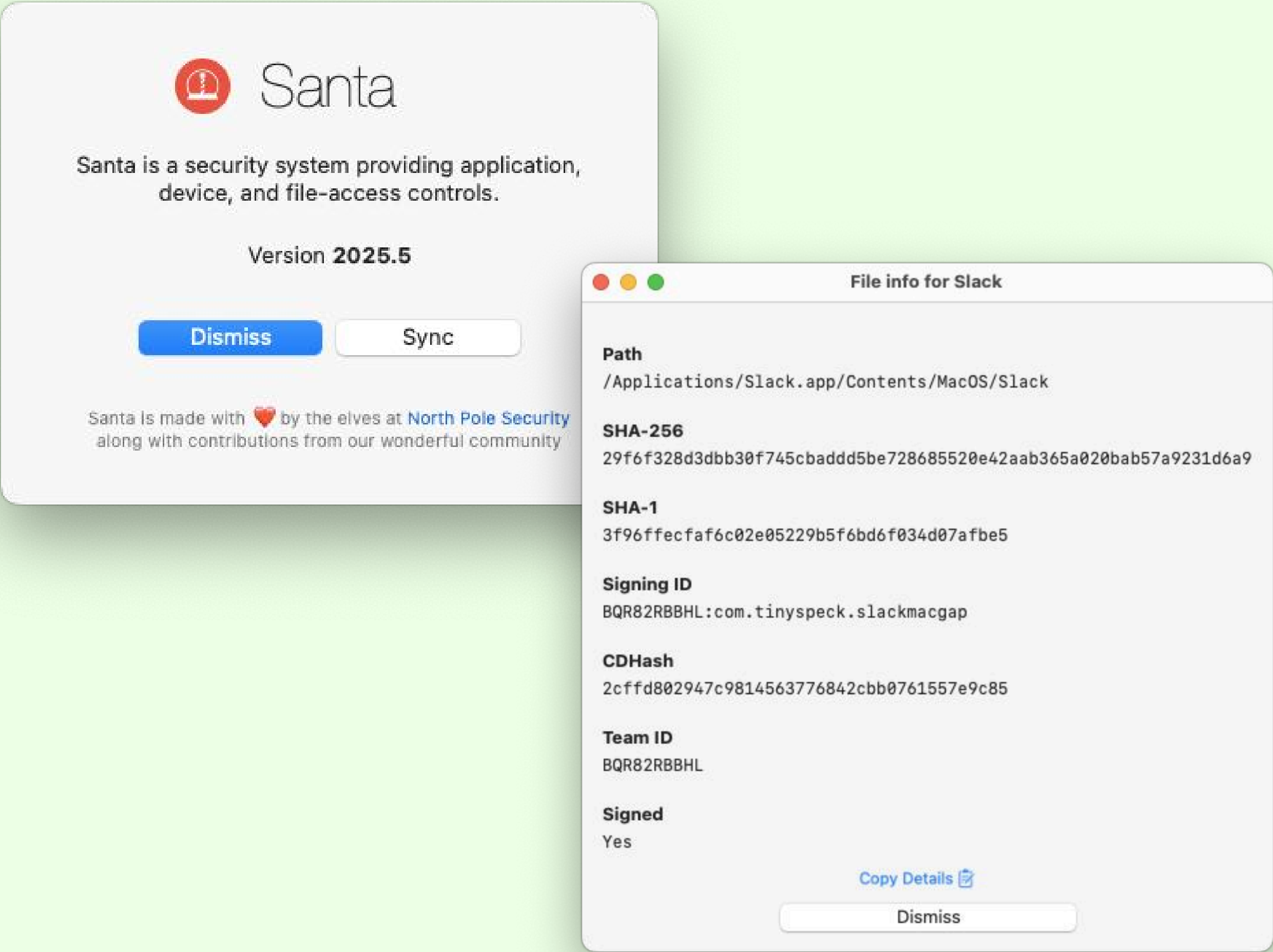

Protecting sudoers config

Should everyone have root?

```
<key>Sudoers</key>
<dict>
  <key>Paths</key>
  <array>
    <dict>
      <key>Path</key>
      <string>/private/etc/sudoers</string>
    </dict>
    <dict>
      <key>Path</key>
      <string>/private/etc/sudoers.d/*</string>
      <key>IsPrefix</key>
      <true/>
    </dict>
    <dict>
      <key>Path</key>
      <string>/private/var/db/sudo/ts/*</string>
      <key>IsPrefix</key>
      <true/>
    </dict>
  </array>
  <key>Options</key>
  <dict>
    <key>AllowReadAccess</key>
    <false/>
    <key>AuditOnly</key>
    <false/>
    <key>RuleType</key>
    <string>PathsWithAllowedProcesses</string>
  </dict>
  <key>Processes</key>
  <array>
    <dict>
      <key>SigningID</key>
      <string>com.apple.sudo</string>
      <key>PlatformBinary</key>
      <true/>
    </dict>
  </array>
</dict>
```

GUI

We've given the UI a major overhaul, added translations, and added usability features for Terminal-averse users.



WHAT IS NEXT?

NEW FEATURES

We've been busy and we're just getting started



SOONER

- CEL Rules
- Telemetry Export



LATER

- FAA Rules in sync protocol
- Tamper Protection Rules
- Re-evaluate running processes



WAY LATER

- Process-Tree Rules
- Enhanced USB Blocking
- ACME mTLS
- Open Telemetry

CEL RULES

Increased control and flexibility

With great power comes great responsibility

```
// Block all executions with GITHUB_TOKEN set  
! has(envs.GITHUB_TOKEN)
```

```
// Prevent using the --inspect flag  
// (useful for Electron apps)  
'--inspect' in args ? BLOCKLIST : ALLOWLIST
```

```
// Only allow running versions of an app signed  
// on or after May 31st 2025.  
target.signing_time >=  
    timestamp('2025-05-31T00:00:00Z')
```


TELEMETRY EXPORT

Santa can produce a tremendous amount of useful telemetry data.

Getting that data off the machine has never been easy.

Until now..

```
{
  "event_time": "2025-06-10T21:08:25.221182781Z",
  "processed_time": "2025-06-10T21:08:25.247378Z",
  "boot_session_uuid":
  "7f39bbb18adb47c9bc3d80cc665dd577",
  "execution": {
    "instigator": {
      "id": {
        "pid": 87417,
        "pidversion": 402687
      },
      "parent_id": {
        "pid": 87328,
        "pidversion": 402538
      },
      "effective_user": {
        "uid": 501,
        "name": "rah"
      },
    },
    "target": {
      "executable": {
        "path": "/usr/bin/yes",
        "truncated": false,
      },
    },
    "decision": "DECISION_DENY",
    "reason": "REASON_SIGNING_ID",
    "mode": "MODE_MONITOR",
  },
}
```

SYNCING FAA RULES

File-access authorization policies require refining over time. This is much easier to do if policies can be distributed with a sync server, like binary authorization rules.

Easy on,
easy off

TAMPER PROTECTION

Santa has tamper protection to prevent existing processes or overzealous users from disabling it.

We're adding the ability to extend this protection to other management tools.

Keep it
all safe

WAY LATER

ENHANCED USB BLOCKING

Blocking USB devices at the device level. Ability to block more types of devices and prevent even raw device writes.

ACME mTLS

Simplified device certificate management for syncing

PROCESS-TREE RULES

Allowing rules to make decisions based on ancestor processes.

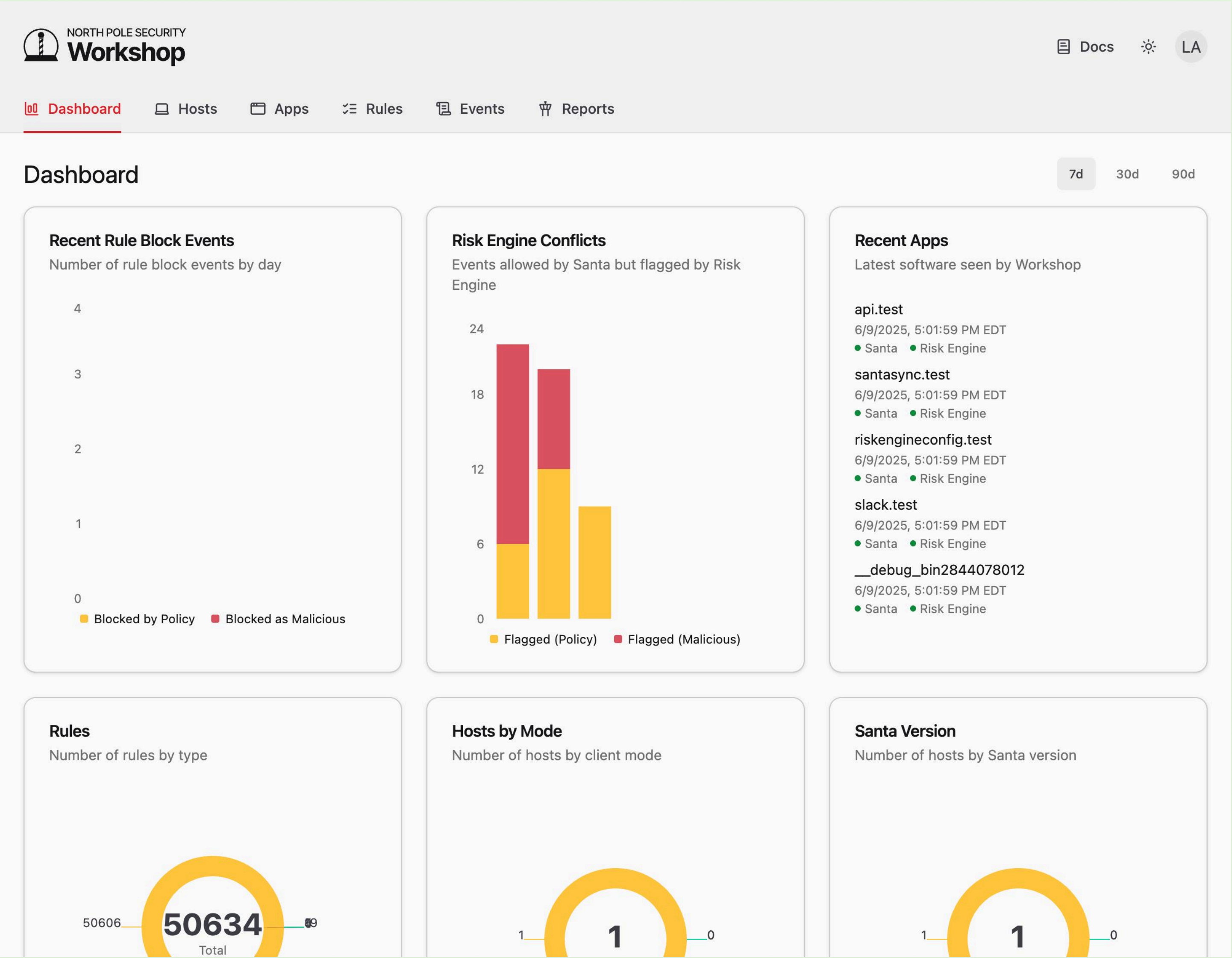
OPEN TELEMETRY

Exporting of Santa metrics to OpenTelemetry.
Resource usage, event counts, cache hit ratios.

INTRODUCING WORKSHOP

WORKSHOP

Making a list, naughty or nice



HOSTS

NORTH POLE SECURITY

Workshop

Docs

LA

Dashboard

Hosts

Apps

Rules

Events

Reports

Hosts

Filter by Hostname...

Filter by Primary User...

+

Last Seen Client Mode

+

OS Version

+

Santa Version

+

SIP Stat

≡

View

Rows per page

10

Page 1 of 1

<<

<

>

>>

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Version 2025.358 (622b64a1)

HOST DETAILS

 NORTH POLE SECURITY

Workshop

Docs

LA

Dashboard

Hosts

Apps

Rules

Events

Reports

Hosts > Host details

Tom-MacBook.localdomain

Sync NowPerform Clean Sync

ID

Primary User

bur@northpole.security (Santa configured) 

Tags

breakglass

postmaster

dev

 (Applied from groups) 

Hardware

MacBookPro18,1

Serial

OS

14.7.5 (23H527)  Update available: 14.7.6

Santa Version

9999.1.1

SIP Status

●

 Enabled

Client Mode

Last Seen: Monitor · Configured: Monitor

Rule Sync Time

May 15, 2025, 02:55 PM

Last Preflight

May 15, 2025, 02:55 PM

Last Postflight

May 15, 2025, 02:55 PM

Host Rules

WORKSHOP

APPS

We keep a full inventory of the apps we've seen

And of course, there's dark mode





Docs

LA

Dashboard

Hosts

Apps

Rules

Events

Reports

Apps

Filter

Sort 1

View

File Name	Vendor	Blockable ID	Santa	Risk Engine	Activity	Host Count	Last Seen
make	59GAB85EFG Apple Inc.	eaeebab8...92a4c095	Allow (35)	-		3	Jun 9, 2025, 05:01 PM
cloud.test	-	da2171ba...f860302f	Allow (3)	Allow		1	Jun 9, 2025, 05:01 PM
covdata	EQHXZ8M8AV Google LLC	f367b049...89548b45	Allow (6)	-		1	Jun 9, 2025, 05:01 PM
workflows.test	-	558986df...3e3fe1d5	Allow (1)	Allow		1	Jun 9, 2025, 05:01 PM
debugserver	59GAB85EFG Apple Inc.	fad3bed0...addb33b4	Allow (3)	-		1	Jun 9, 2025, 05:01 PM
santasync.test	-	a1182ff1...be685afa	Allow (1)	Allow		1	Jun 9, 2025, 05:01 PM
utils.test	-	1e7008d5...20589cc8	Allow (1)	Allow		1	Jun 9, 2025, 05:01 PM
risk_engine.test	-	400cfb7e...6d0d6020	Allow (1)	Allow		1	Jun 9, 2025, 05:01 PM
exceptions.test	-	5ecba20f...064f0995	Allow (1)	Allow		1	Jun 9, 2025, 05:01 PM

WORKSHOP

RULES

 NORTH POLE SECURITY

Workshop

Docs



Dashboard

Hosts

Apps

Rules

Events

Reports

API Keys

Audit Events

Settings

Rules

Create Rule

Filter

Sort

View

Identifier	Tag	Comment	Rule Type	Policy	Actions
U68MSDN6DR:org.whisper... Signal Messenger, LLC	host:7228546F...ACE830BE	Rule(s) added by self-service approval workflow from tag: dev (Application: Signal)	Signing ID	 Allowlist	...
U68MSDN6DR:org.whisper... Signal Messenger, LLC	host:7228546F...ACE830BE	Rule(s) added by self-service approval workflow from tag: dev (Application: Signal)	Signing ID	 Allowlist	...
U68MSDN6DR:org.whisper... Signal Messenger, LLC	host:7228546F...ACE830BE	Rule(s) added by self-service approval workflow from tag: dev (Application: Signal)	Signing ID	 Allowlist	...
U68MSDN6DR:chrome_cras... Signal Messenger, LLC	host:7228546F...ACE830BE	Rule(s) added by self-service approval workflow from tag: dev (Application: Signal)	Signing ID	 Allowlist	...
U68MSDN6DR:org.whisper... Signal Messenger, LLC	host:7228546F...ACE830BE	Rule(s) added by self-service approval workflow from tag: dev (Application: Signal)	Signing ID	 Allowlist	...

SLACK

Slack integration lets you handle approval workflows without opening the web UI



North Pole Security

APP 4:38 PM

The following software was prevented from executing and requires your approval

Name:

IMDMessageServicesAgent

When:

2025-02-12 15:21:39

Where:

ragged.northpole.security

Publisher:

Apple Inc.

User:

appleseed

Allowed by Risk Engine?

All Checks Passed

Approve

Report Malware

Details

Dismiss



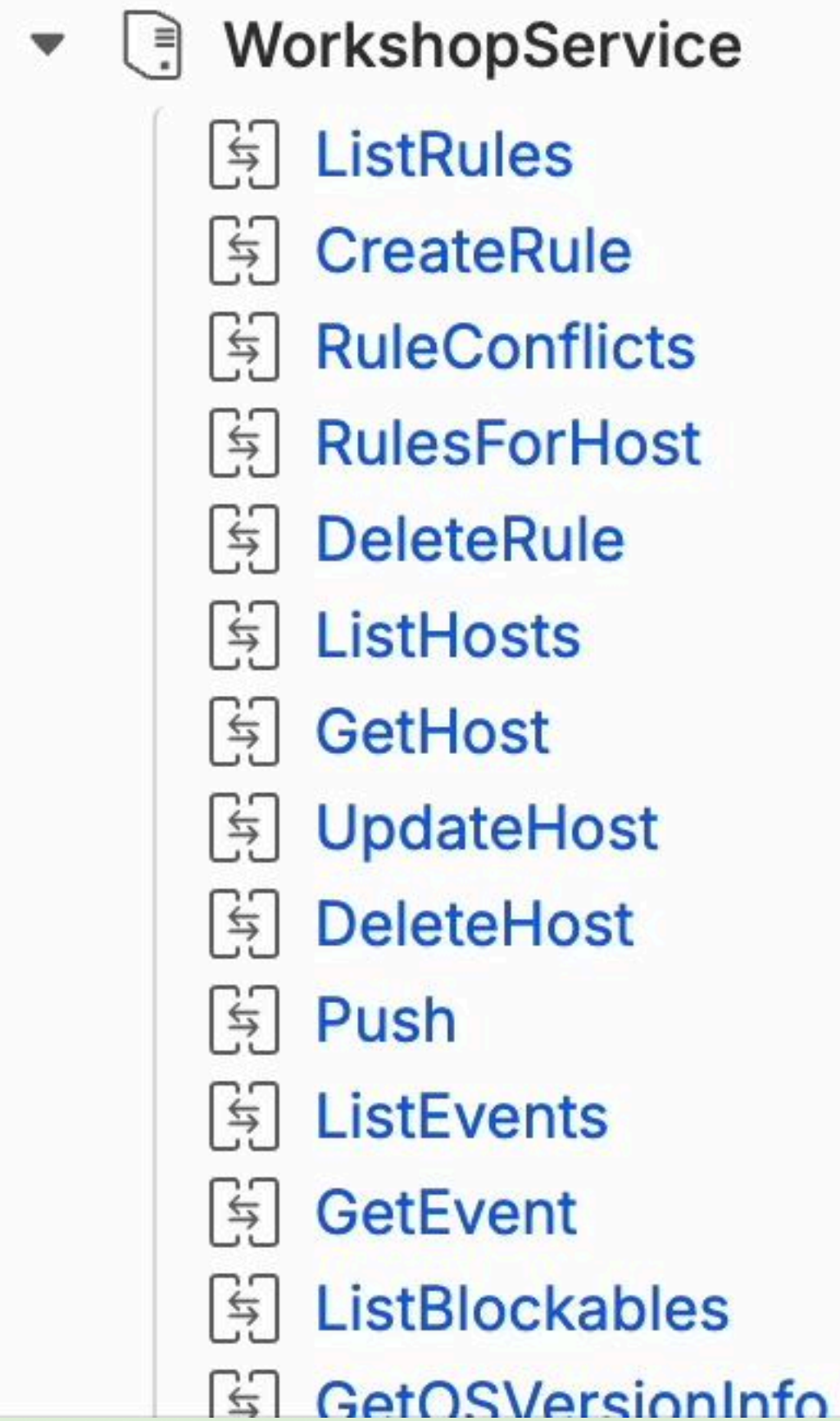
NORTH POLE
SECURITY

SANTA IN THE SUMMER

API

Everything the UI can do

The API can do better



A screenshot of a REST client interface showing a list of API endpoints for a service named 'WorkshopService'. The endpoints are listed with their corresponding HTTP methods (indicated by icons) and names. The endpoints are: ListRules, CreateRule, RuleConflicts, RulesForHost, DeleteRule, ListHosts, GetHost, UpdateHost, DeleteHost, Push, ListEvents, GetEvent, ListBlockables, and GetOSVersionInfo.

- ▼ WorkshopService
 - ↔ ListRules
 - ↔ CreateRule
 - ↔ RuleConflicts
 - ↔ RulesForHost
 - ↔ DeleteRule
 - ↔ ListHosts
 - ↔ GetHost
 - ↔ UpdateHost
 - ↔ DeleteHost
 - ↔ Push
 - ↔ ListEvents
 - ↔ GetEvent
 - ↔ ListBlockables
 - ↔ GetOSVersionInfo

A faint, stylized illustration of Santa Claus in a light red color, centered in the background. He is wearing a hat with a pom-pom, a suit with a belt, and has a large, bushy beard. His arms are slightly out to the sides.

Demos

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A faint, stylized illustration of Santa Claus is visible in the background. He is wearing his iconic red hat with a white pom-pom, a red suit with white trim, and has a white beard. His hands are visible at the sides, holding what appear to be reindeer antlers.

Thank you!

Get on the nice list
info@northpole.security

northpole.security